

Trapped-Ion Page-Wootters Experiment: Protocol for Testing Ultrametricity

| 2026-07-01

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Abstract

We propose a trapped-ion experiment to test the Sufficient Condition Theorem: that diagonal clock-rest coupling $\hat{H}_{CR}$ in the clock eigenbasis produces exact ultrametric conditional state overlaps ($UVR = 0$) in the electronic states as the clock and M motional Fock states as the rest system. Laser-driven carrier and sideband transitions provide $\hat{H}_{CR}$ Extra close brace or missing open brace. Conditional state tomography at multiple clock readings yields the overlap matrix O_{ij} .

1. Physical Implementation

1.1 System

Component	Physical Realization	Parameters (Yb ⁺)
Clock $\mathcal{H}_C$	$N = 4$ Zeeman sublevels of $^2S_{1/2} F=0, m_F\rangle$	$\Delta E \approx 1\text{--}10$ MHz (B-field tunable)
Rest $\mathcal{H}_R$	Axial motional mode (harmonic oscillator)	$\omega_z/2\pi \approx 1\text{--}2$ MHz
Coupling $\hat{H}_{CR}$	Laser-ion interaction (carrier + sidebands)	Rabi frequency $\Omega/2\pi \approx 10\text{--}100$ kHz

1.2 Hamiltonian

$$\begin{aligned} \hat{H} &= \hat{H}_C \otimes \hat{I}_R + \hat{I}_C \otimes \hat{H}_R + \hat{H} \\ &\end{aligned}$$

where:

$$\begin{aligned} \hat{H}_C &= \sum_k E_k |e_k\rangle\langle e_k|, \quad \hat{I}_C = \sum_k |e_k\rangle\langle e_k| \\ \hat{H}_R &= \hbar\omega_z \hat{a}^\dagger \hat{a} \end{aligned}$$

1.3 Clock-Rest Interaction Engineering

The laser-ion interaction in the Lamb-Dicke regime:

$$\hat{H}_{CR} = \sum_k \Omega_{kl} |e_k\rangle\langle e_l| \otimes \hat{V}_{kl} + \text{h.c.}$$

where $\hat{V}_{kl}$ acts on the motional space.

Diagonal coupling (predicted UVR = 0%):

- Use **carrier transitions only**: $\hat{V}_{kk} = \hat{I}_R - \text{Laser drives } |e_k\rangle\langle e_l| \rightarrow |e_l\rangle\langle e_l| \text{ without changing motional state}$ – $\hat{H}$ is strictly diagonal in the clock eigenbasis
- Experimental check: motional state unchanged after clock-state transfer

Nondiagonal coupling (predicted UVR ≈ 32%):

- Use **first-order red/blue sidebands**: $\hat{V}_{kl} = \eta(\hat{a} + \hat{a}^\dagger) - \text{Laser drives } |e_k\rangle\langle n| \rightarrow |e_l\rangle\langle n \pm 1|$ – $\hat{H}$ has off-diagonal elements connecting different motional eigenstates
- Experimental check: motional state changes with clock-state transfer

Key tunability: By tuning laser frequency to carrier vs. sideband, we switch between the two regimes in the same apparatus. This is the experiment's central advantage.

2. WDW State Preparation

2.1 The Constraint

The WDW state $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_R$ satisfies:

$$(\hat{H}_C \otimes \hat{I}_R + \hat{I}_C \otimes \hat{H}_R + \hat{H}) |\Psi\rangle = 0$$

2.2 Preparation Protocol

We use **adiabatic state preparation**:

1. **Initialize:** Prepare the system in a product state $|e_0\rangle \otimes |0\rangle$ (ground clock state, motional ground state)
2. **Ramp coupling:** Slowly turn on $\hat{H}_{CR}$ from 0 to target strength, keeping the total Hamiltonian at each step
3. **Verify:** The final state is an eigenstate of $\hat{H}$ with eigenvalue E_0

Energy shift for zero eigenvalue. The initial energy is $E_0^{(0)} = E_0 + 0 = E_0$. To make the WDW eigenvalue zero, we shift the total Hamiltonian:

$$\hat{H}_{\text{shifted}} = \hat{H} - E_0 \hat{I}$$

After adiabatic evolution, the state $|\Psi\rangle$ satisfies $\hat{H}_{\text{shifted}} |\Psi\rangle = 0$ — the WDW constraint.

Adiabatic condition: Ramp time $\tau \gg \max_{n \neq 0} \hbar / |E_n - E_0| \approx 2\pi / \min(\Omega, \omega_z)$

For $\Omega/2\pi = 50$ kHz: $\tau \gtrsim 20 \mu\text{s}$ (easily achievable; ion coherence times are ~ 100 ms)

2.3 Validation

After preparation, verify:

1. $\langle \Psi | \hat{H} | \Psi \rangle \approx 0$ (zero expectation value)
2. $\langle \Psi | \hat{H}^2 | \Psi \rangle \approx 0$ (zero variance — genuine eigenstate)
3. Repeat state tomography to verify preparation fidelity

3. Conditional State Measurement

3.1 Conceptual Protocol

The conditional state at "clock time" k is:

$$|\psi(t_k)\rangle = \frac{\langle e_k | \Psi \rangle}{|\langle e_k | \Psi \rangle|}$$

where t_k parameterizes the clock reading (not physical time — it's the eigenvalue label E_k).

3.2 Experimental Sequence

For each clock index $k = 0, \dots, N - 1$:

1. **Prepare WDW state** $|\Psi\rangle!$ (adiabatic protocol from §2.2)
2. **Projectively measure clock:** Measure in $|e_k\rangle$ basis $\rightarrow$ post-select on outcome k
3. **Tomography of rest state:** The post-selected motional state is $|\psi(t_k)\rangle$
4. **Reconstruct density matrix:** ρ_k via motional state tomography (displacement operations + parity measurements)

Repeats: $\sim 10,000$ per clock index for statistical convergence (comparable to recent ion tomography experiments).

3.3 Motional State Tomography

For $M = 4$ motional Fock states (truncated):

1. Apply displacement $D(\alpha)$ with varying α
2. Measure parity $\hat{P} = e^{i\pi\hat{a}^\dagger\hat{a}}$ of the motional state
3. Reconstruct Wigner function $W(\alpha)$
4. Extract density matrix ρ_k via inverse Radon transform

Measurement time: ~ 1 ms per shot $\times 10,000$ shots $\times 4$ clock states = 40 seconds per data point.

4. Ultrametricity Test

4.1 Overlap Matrix

From reconstructed ρ_k , compute:

$$O_{ij} = |\langle\psi(t_i)|\psi(t_j)\rangle| = \sqrt{\text{Tr}(\rho_i\rho_j)}$$

This requires $N(N - 1)/2 = 6$ pairwise overlaps for $N = 4$.

4.2 Parisi Ultrametricity Violation Rate (UVR)

For each triangle (i, j, k) with $i < j < k$:

1. Sort overlaps: $O_{(1)} \leq O_{(2)} \leq O_{(3)}$
2. Test: $O_{(2)} = O_{(3)}$ (strong triangle inequality in sorted form)
3. Count violations within tolerance ϵ

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4.3 Predicted Outcomes

Regime	$\hat{H}_{CR}$ Type	Predicted UVR (theory)	Predicted UVR (exp., $\epsilon = 0.03$)
Diagonal	Carrier only	0.00%	< 5% (limited by tomography noise)
Nondiagonal	Sideband coupling	32 ± 3%	28–37% (statistical + systematic)

Statistical significance: With $N = 4$ (4 triangles), the binomial probability of observing 0 violations in the diagonal regime when the true UVR is 32% is $p = (1 - 0.32)^4 \approx 0.21$ — insufficient. We need $N \geq 5$ (10 triangles): $p = (1 - 0.32)^{10} \approx 0.02$.

Recommendation: $N = 6$ **clock states** (20 triangles, $p < 0.001$ for false negative).

5. Experimental Requirements

5.1 Required Capabilities

Requirement	Value	Status
Single ion trapping (Yb ⁺ or Ca ⁺)	Standard Paul trap	✓ Established [established]
$N \geq 6$ resolvable Zeeman sublevels	B-field $\approx 5\text{--}10$ G	✓ Routine [established]
Motional ground state cooling	$\bar{n} < 0.1$	✓ Routine (sideband cooling) [established]
Individual addressing of transitions	Multiple AOM channels	✓ Routine [established]
Motional state tomography ($M \geq 4$)	Displacement + parity	✓ Demonstrated (Leibfried et al., 1996; Lv et al., 2018) [established]
Adiabatic state preparation	Ramped laser intensity	✓ Routine [established]
Coherence time > 100 ms	Magnetic shielding	✓ Achievable [established]

5.2 Clock Spectrum Engineering

For p-adic spectrum testing, the Zeeman sublevel energies must follow:

$$E_k = E_0 \cdot p^k \quad (k = 0, 1, \dots, N - 1)$$

This is achieved via a **spatially varying magnetic field** (quadrupole gradient) so the Zeeman shift varies across sublevels. With $N = 6$ and $p = 2$: $E_k \propto (1, 2, 4, 8, 16, 32)$ — requiring a factor of 32 in energy range, feasible with ~ 50 G field gradient over $\sim 1\ \mu\text{m}$ ion displacement.

Simplified spectrum (equal spacing): If p-adic spectrum engineering proves challenging, use equally spaced Zeeman sublevels as baseline. The UVR is predicted to be similar (32–35%) since the violation rate depends on coupling structure, not clock spectrum details — confirmed by the 8000-trial computational search.

6. Noise Analysis and Error Budget

6.1 Sources of Systematic Error

Error Source	Magnitude	Effect on UVR
Tomography reconstruction noise	$\Delta O_{ij} \approx 0.02\text{--}0.05$	Falsely elevates UVR by $\sim 3\text{--}8\%$
Incomplete adiabatic preparation	Fidelity < 0.99	Mixes ideal WDW state with excited states
Motional heating ($\dot{\vec{n}}$)	~ 10 quanta/s	Decoherence during tomography
Magnetic field fluctuations	$\Delta B/B \sim 10^{-4}$	Dephasing of clock states
Laser intensity noise	$\Delta\Omega/\Omega \sim 10^{-3}$	Fluctuations in $\hat{H}_{CR}$

6.2 Statistical Error

With $N = 6$ clock states (20 triangles) and $R = 10^4$ repetitions:

- Per-triangle binomial error on UVR: $\sigma_{\text{UVR}} \approx \sqrt{p(1-p)/R_{\text{tri}}}$
- For $p = 0.32$ and 333 effective trials per triangle: $\sigma_{\text{UVR}} \approx 0.026$
- Total UVR uncertainty: $\approx 3\%$

6.3 Distinguishing Diagonal from Nondiagonal

The effect size is 32% (separation between diagonal UVR $\approx 0\%$ and nondiagonal UVR $\approx 32\%$). Required SNR > 5 : need $\sigma_{\text{UVR}} < 6.4\%$, which is satisfied by tomography noise of 2–5%.

The experiment is feasible with current technology. [established]

7. Timeline and Resource Estimate

Phase	Duration	Activity
1. Setup	2 weeks	Trap loading, cooling calibration, magnetic field tuning
2. Spectrum calibration	1 week	Measure Zeeman sublevel energies, calibrate addressing
3. Diagonal regime	2 weeks	Carrier-only coupling, WDW preparation, tomography, data taking
4. Nondiagonal regime	2 weeks	Sideband coupling, same protocol
5. Analysis	1 week	Overlap matrix reconstruction, UVR computation, systematics

Total: 8 weeks on an existing trapped-ion apparatus.

8. Connection to CMB and Bruhat-Tits

If the experiment confirms:

- **Diagonal regime** $\rightarrow$ **UVR $\approx$ 0%**: Validates the sufficient condition theorem experimentally
- **Nondiagonal regime** $\rightarrow$ **UVR $\approx$ 32%**: Confirms the universality of the violation rate

Then the Sufficient Condition Theorem is experimentally verified, completing the physical foundations of the Bridge Theorem. The CMB null result is then interpreted as evidence that early-universe clock-rest coupling was diagonal — the "classical ideal clock" regime.

Open question for future work: Can the trapped-ion platform simulate a p-adic clock spectrum and test the conjecture that ultrametricity is equivalent to the p-adic tree structure of Bruhat-Tits buildings?

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